



# Law of Sines

Trigonometry Worksheet · Grade 10-12

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Score:     / 9

## Learning Objectives

- Apply the Law of Sines to find missing sides of oblique triangles
- Apply the Law of Sines to find missing angles of oblique triangles
- Correctly label angles and their corresponding opposite sides

Use the Law of Sines to find each missing side or angle, rounding answers to two decimal places where needed.

**1. In triangle ABC, angle A = 110°, a = 125 in, b = 100 in. Find angle B.**

$$\frac{125}{\sin 110^\circ} = \frac{100}{\sin B}$$

Answer: \_\_\_\_\_

**2. In triangle ABC, angle A = 110° and angle B = 48.74°. Find angle C.**

$$C = 180^\circ - A - B$$

Answer: \_\_\_\_\_

**3. In triangle ABC, angle A = 110°, angle C = 21.26°, and a = 125 in. Find side c.**

$$\frac{125}{\sin 110^\circ} = \frac{c}{\sin 21.26^\circ}$$

Answer: \_\_\_\_\_

**4. In triangle ABC, angle A = 40°, angle B = 75°, a = 12 cm. Find side b.**

$$\frac{12}{\sin 40^\circ} = \frac{b}{\sin 75^\circ}$$

Answer: \_\_\_\_\_

**5. In triangle ABC, angle A = 40°, angle B = 75°. Find angle C.**

$$C = 180^\circ - 40^\circ - 75^\circ$$

Answer: \_\_\_\_\_

**6. In triangle ABC, angle A = 35°, angle C = 80°, a = 10 m. Find side c.**

$$\frac{10}{\sin 35^\circ} = \frac{c}{\sin 80^\circ}$$

Answer: \_\_\_\_\_



7. In triangle ABC,  $a = 8$ ,  $b = 11$ , angle  $A = 32^\circ$ . Find angle B (acute solution).

$$\frac{8}{\sin 32^\circ} = \frac{11}{\sin B}$$

Answer: \_\_\_\_\_

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8. In triangle ABC, angle  $A = 25^\circ$ , angle  $B = 60^\circ$ ,  $a = 7$  ft. Find side b.

$$\frac{7}{\sin 25^\circ} = \frac{b}{\sin 60^\circ}$$

Answer: \_\_\_\_\_

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9. In triangle ABC, angle  $B = 95^\circ$ , angle  $C = 30^\circ$ ,  $b = 20$  in. Find side c.

$$\frac{20}{\sin 95^\circ} = \frac{c}{\sin 30^\circ}$$

Answer: \_\_\_\_\_





Remind students that the side opposite an angle shares the same letter (lowercase) and that calculators must be set to degree mode.

## Solutions

1. In triangle ABC, angle A =  $110^\circ$ , a = 125 in, b = 100 in. Find angle B.

$$\frac{125}{\sin 110^\circ} = \frac{100}{\sin B}$$

- Set up the Law of Sines using the pair a/sin A and b/sin B
- Cross multiply to get sin B equals 100 times sin 110 degrees divided by 125
- Compute sin B is approximately 0.7518
- Take the inverse sine to find B is approximately 48.74 degrees

**Answer:**  $B \approx 48.74^\circ$

2. In triangle ABC, angle A =  $110^\circ$  and angle B =  $48.74^\circ$ . Find angle C.

$$C = 180^\circ - A - B$$

- Recall that the three interior angles of a triangle sum to 180 degrees
- Subtract angle A and angle B from 180 degrees
- C is approximately 180 minus 110 minus 48.74, which is 21.26 degrees

**Answer:**  $C \approx 21.26^\circ$

3. In triangle ABC, angle A =  $110^\circ$ , angle C =  $21.26^\circ$ , and a = 125 in. Find side c.

$$\frac{125}{\sin 110^\circ} = \frac{c}{\sin 21.26^\circ}$$

- Set up the Law of Sines pairing side a with angle A and side c with angle C
- Solve for c by multiplying both sides by sin 21.26 degrees
- Compute c equals 125 times sin 21.26 degrees divided by sin 110 degrees
- Side c is approximately 48.22 inches

**Answer:**  $c \approx 48.22$  in

4. In triangle ABC, angle A =  $40^\circ$ , angle B =  $75^\circ$ , a = 12 cm. Find side b.

$$\frac{12}{\sin 40^\circ} = \frac{b}{\sin 75^\circ}$$

- Use the Law of Sines pairing side a with angle A and side b with angle B
- Multiply both sides by sin 75 degrees
- Compute b equals 12 times sin 75 degrees divided by sin 40 degrees
- Side b is approximately 18.03 centimeters

**Answer:**  $b \approx 18.03$  cm



5. In triangle ABC, angle A = 40°, angle B = 75°. Find angle C.

$$C = 180^\circ - 40^\circ - 75^\circ$$

- Use the fact that the interior angles sum to 180 degrees
- Subtract the two known angles from 180
- C equals 65 degrees

**Answer:**  $C = 65^\circ$

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6. In triangle ABC, angle A = 35°, angle C = 80°, a = 10 m. Find side c.

$$\frac{10}{\sin 35^\circ} = \frac{c}{\sin 80^\circ}$$

- Apply the Law of Sines pairing side a with angle A and side c with angle C
- Solve for c by multiplying both sides by sin 80 degrees
- Compute c equals 10 times sin 80 degrees divided by sin 35 degrees
- Side c is approximately 17.17 meters

**Answer:**  $c \approx 17.17 \text{ m}$

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7. In triangle ABC, a = 8, b = 11, angle A = 32°. Find angle B (acute solution).

$$\frac{8}{\sin 32^\circ} = \frac{11}{\sin B}$$

- Set up the Law of Sines pairing a with angle A and b with angle B
- Cross multiply to get sin B equals 11 times sin 32 degrees divided by 8
- Compute sin B is approximately 0.7289
- Take the inverse sine to obtain B approximately 46.84 degrees

**Answer:**  $B \approx 46.84^\circ$

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8. In triangle ABC, angle A = 25°, angle B = 60°, a = 7 ft. Find side b.

$$\frac{7}{\sin 25^\circ} = \frac{b}{\sin 60^\circ}$$

- Use the Law of Sines pairing side a with angle A and side b with angle B
- Multiply both sides by sin 60 degrees
- Compute b equals 7 times sin 60 degrees divided by sin 25 degrees
- Side b is approximately 14.34 feet

**Answer:**  $b \approx 14.34 \text{ ft}$

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9. In triangle ABC, angle B = 95°, angle C = 30°, b = 20 in. Find side c.

$$\frac{20}{\sin 95^\circ} = \frac{c}{\sin 30^\circ}$$

- Apply the Law of Sines pairing side b with angle B and side c with angle C
- Solve for c by multiplying both sides by sin 30 degrees
- Compute c equals 20 times sin 30 degrees divided by sin 95 degrees
- Side c is approximately 10.04 inches

**Answer:**  $c \approx 10.04 \text{ in}$

